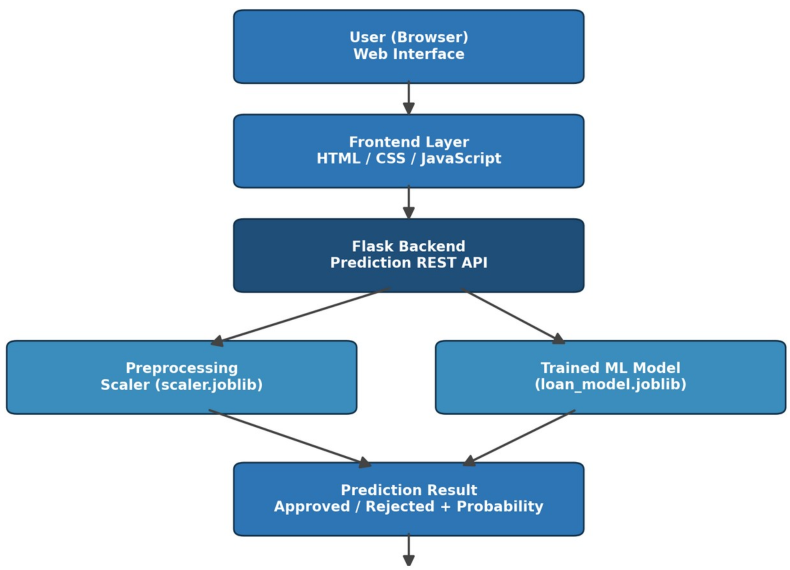
Phase 3: Project Design Phase

System Architecture Diagram

Smart Lender follows a simple, layered client-server architecture. The user interacts with a web-based frontend, which sends the applicant's details to a Flask backend through a REST API. The backend preprocesses the input using a saved StandardScaler and passes it to the trained machine learning model, which returns the prediction. The diagram below illustrates the flow of data through the application, from the user's browser to the final prediction result.



User Interface Mockups

The web interface is designed with a clean, single-page, FinTech-inspired layout so that an applicant can enter their details and receive a prediction without any page reloads. The planned screens/sections are as follows:

- Header Section: Application name/logo and a short tagline describing the purpose of the tool.
- Applicant Details Form: Input fields for number of dependents, education, employment status, annual income, loan amount, loan term, CIBIL score, and asset values (residential, commercial, luxury, and bank assets).

- Predict Button: Submits the form data to the /predict API endpoint via JavaScript (asynchronous request, no page reload).
- Result Panel: Displays the prediction outcome (Approved / Rejected) along with the approval probability, using clear color coding (e.g., green for approval, red for rejection) for quick readability.

Actual screenshots of the implemented interface, captured after running the application, are included in the Project Demonstration phase (Folder 8).